

Group Project Assignment 2: Data Modeling

1. Scope of Societal Problem

Given the societal problem chosen in Assignment 1, that being, the housing crisis in the Netherlands with a focus on Amsterdam, we find that a database that accrues a wide variety of available housing options that is available to the public would ease housing search for both residents of the Netherlands and newcomers. Despite their existing services such as Pararius and Huurwoningen which gather data from many housing corporations, it still misses a large amount of housing listings due to corporations themselves making the decisions for where they list their assets, so there is no centralised database and people have to gather multiple different housing services and corporations to be able to find housing.

2. How data would interact

- a. There are many types of landlords (private, housing corporation, etc)
- b. Landlords can hold many listings
- c. Landlords can add or remove listings

3. Entities

- a. Listing
- b. Landlord
- c. LandlordType
- d. Property
- e. Location
- f. HousingType
- g. PricingType

4. Relationships

- a. Landlord manages Listing → one-to-many (1 landlord : many listings)
- b. Landlord is a LandlordType → many-to-one (many landlords : 1 landlord type)
- c. Listing is for Property → many-to-one (many listings : 1 property)
- d. Listing has a PricingType → many-to-one (many listings : 1 pricing type)
- e. Listing has a RentalStatus → many-to-one (many listings : 1 rental status)
- f. Property is in Location → one-to-one (one property : 1 location)
- g. Property is a HousingType → many-to-one (many properties : 1 housing type)

Normalization:

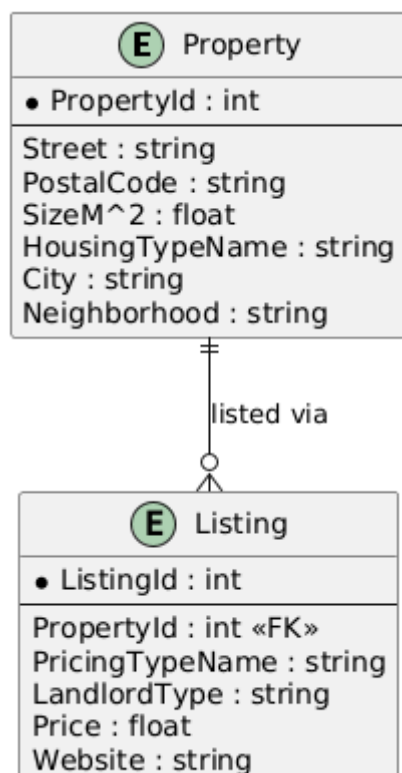
A. 1NF



Here, every row has only one listing, and all the characteristics of each listing are divided into their own cells, the type of property (room, flat, house etc), the type of price, the price itself, all the elements of the address, the site where the renter can see the listing, the surface area, what type of landlord, and the property id.

The primary key is the ListingID.

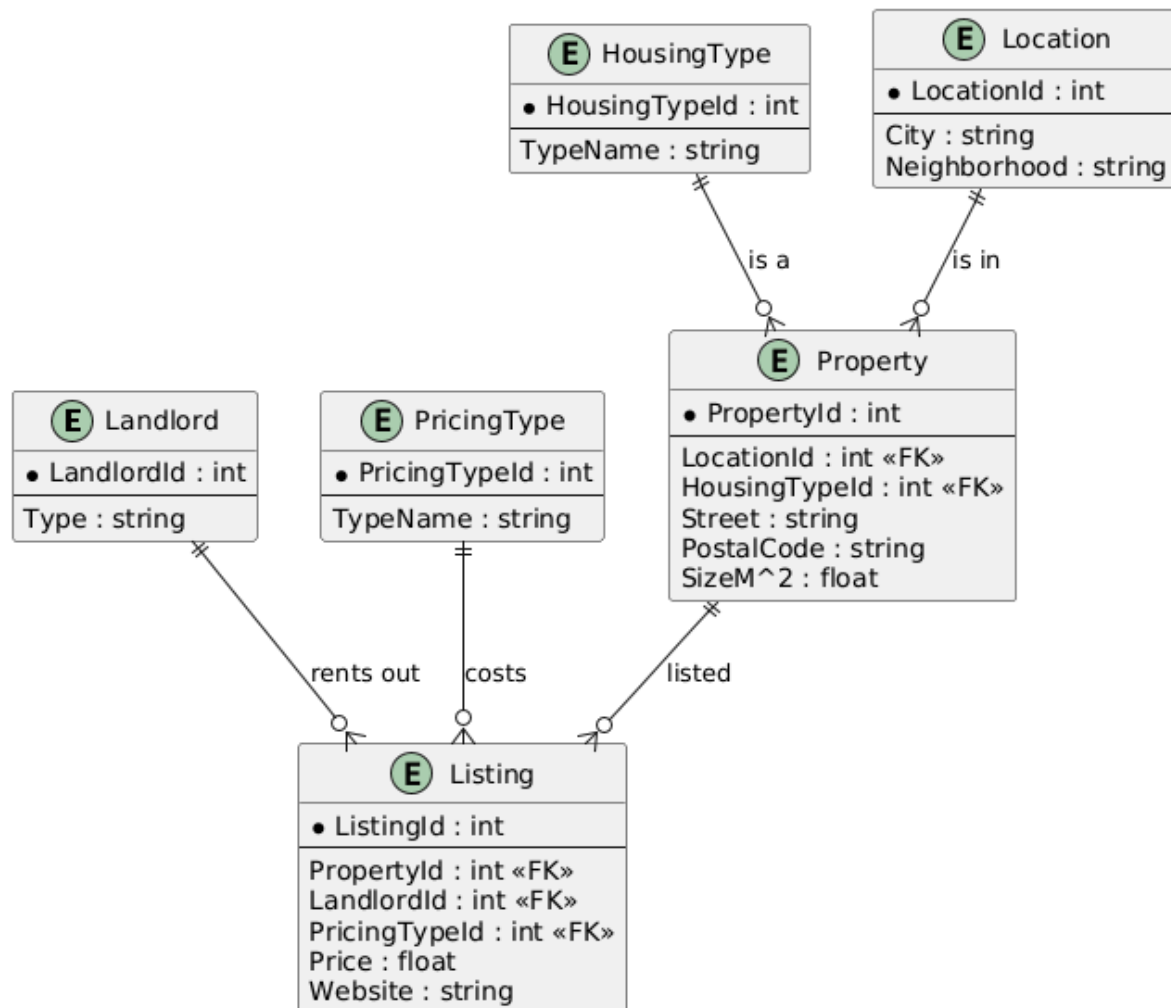
B. 2NF



Here, the database is split into two tables: Properties (with PropertyId as the primary key) and Listings, with ListingId as the primary key. The original

Listing table mixed attributes describing the property (street, postal code, size, housing type, city, neighborhood) with attributes describing the listing itself (pricing type, price, landlord type, website). Since the property attributes depend only on PropertyId and not on the whole listing record, they would be duplicated every time a property was relisted. To satisfy 2NF, we moved them into a separate Property table, linked to Listing through a PropertyId foreign key.

C. 3NF



The 2NF tables still stored descriptive values such as housing type, pricing type, landlord type, and city and neighborhood directly in the Property and Listing tables, repeating them across many rows. Some of these also depended on other non-key attributes rather than on the primary key alone, for example a neighborhood being tied to its city. To fix this, we moved them into their own lookup tables (HousingType, PricingType, Landlord, and Location) and referenced them through foreign keys. Each piece of information is now stored only once, which keeps the data consistent and easier to update.